Engineering - Undergraduate Level Assessment 15 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. f(X) = $[\pi(1 + X^2)]^-1 - \infty < x < \infty. If Y$ =X^2, what is the density function of Y ?
   (a) h(y) = [2 / $\{\pi(1 + \sqrt{y})\}$] for y > 0 and = 0 otherwise
   (b) h(y) = [2 / $\{\pi$(1 + y)$\}$]y > 0 and = 0 otherwise
   (c) h(y) = [1 / $\{\pi(1 + y^2)\}$] for y > 0 and = 0 otherwise
   (d) h(y) = [1 / $\{2\pi(1 + \sqrt{y})\}$] for y > 0 and = 0 otherwise
   (e) h(y) = [1 / $\{2\pi(1 + y^2)\}$] for y > 0 and = 0 otherwise

2. Find the current through a 0.01 capacitor at t = 0 if the voltage across it is (a) 2 sin 2π 10^6 t V; (b) - 2 e$^-(10)7tV; (c) - 2e - (10)7tsin2\pi10^6$ t V.
   (a) 0.1256 A, 0.2 A, 0.1256 A
   (b) 0.1256 A, 0.3 A, 0.1256 A
   (c) 0.2345 A, 0.1 A, - 0.4567 A
   (d) 0.2 A, 0.1 A, 0.2 A
   (e) 0 A, 0.1 A, 0 A

3. A certain load to be driven at 1750 r/min requires a torque of 60 lb. ft. What horsepower will be required to drive the load ?
   (a) 15 hp
   (b) 20 hp
   (c) 10 hp
   (d) 35 hp
   (e) 5 hp

4. The moving coil-meters, damping is provided by
   (a) the resistance of the coil.
   (b) the inertia of the coil.
   (c) the coil spring attached to the moving.
   (d) the electric charge in the coil.
   (e) the magnetic field created by the coil.

5. The current through a resistor of 2Ω is given by i(t) = cos^2. Find the energy dissipated in the resistor from t_0 = 0 to t_1 = 5 sec.
   (a) 5.75 J

(b) 7.50 J

(c) 4.75 J

(d) 1.25 J

(e) 3.75 J

True / False (10 marks)

Answer True or False

6. True or False: The radiation resistance of a single-turn circular loop with a circumference of (1/4) wavelength is 0.77 ohm.

7. True or False: One liter-atmosphere of work is equivalent to 106.4 joules.

8. True or False: Newton's Law of cooling for convective heat transfer is represented by the equation q = hAT.

9. True or False: A D-flip-flop is transparent when the output is triggered by the rising edge of the clock signal.

10. True or False: The electrostatic energy of a point charge q inside a conducting shell is zero due to shielding effects of the shell.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the principles involved in introducing air through a nozzle into water to create bubbles of 2 mm diameter. Calculate the necessary pressure difference between the air at the nozzle and the surrounding water, considering the surface tension of water given as $= 72.7 \times 10^{-}3Nm^{-}1$.

11. Analyze the function F(s) = (3 s + 1) / {(s - 1)(s^2 + 1)} and determine its inverse Laplace transform. Explain your reasoning and the steps taken in your analysis.

End of Paper